Jhon Ramirez

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EDUCATION

University at Buffalo **Buffalo, NY**
Electrical Engineering B.S May 25. Expected Graduation: Dec. 2026
Robotics And Automation M.S
Relevant Coursework: Programming in C and C++. Signals and systems, Digital principles, Circuit Analysis, Electronic Devices and Circuits, Applied Probability, Communication systems, Embedded systems, Energy systems, HDL programmable logic, Wearable and Implantable Sensors, Power Electronics, Mechatronics, Robotics algorithms, Intro Machine learning.

SKILLS

Intermediate with the use of Fusion 360, Multisim,
Proficient in Spanish speaking and writing
Intermediate in programing languages: C/C++, MATLAB, VHDL, ROS, python
Proficient with soldering Through holes and surface mounted devices, proficient in machine tools

PROJECTS

Kinetic sand table *June 2021- Jan 2024*

- Using a motherboard from a 3D printer modified to be able to work with driving the two stepper motors, that make the Y and X axis.
- Applied a Raspberry Pi 3A that has Klipper and Mainsail server flashed on to be able send information to the machine, along with any troubleshooting errors.
- The CNC moves a magnet that pulls a metal ball above it making all the drawings and designs that were imported into the machine.

Magic Mirror *April 20, 2022*

- Using a Raspberry Pi 5, a Pi Camera, a Microphone, and a display. The mirror will be able to display weather updates, email, calendar and more.
- This project was inspired by a project done by a community that has already done this before, with the codes and instructions given on their GitHub page.
- The Mirror that I made includes face ID, google assistance with the use of both the camera running on python and Face-recognition, and a USB microphone.

8x8x8 LED Cube *Dec. 2021- April. 2026*

- Used 5mm Led diodes to create a 8x8x8 led matrix cube being controlled by an ESP32.
- Created a PCB design for the control of the LED including 74HC595 and ULN2803, a buck boost converter and the ESP32.
- Learned how to create a web app to be able to create as well as control the animation displayed on the cube.

RELEVANT EXPERIENCE

CytoCybernetics **Buffalo, NY**
Electrical engineer assistant/ Intern *June 2023 – Aug.2023*

- Designed a simple Printed Circuit Board (PCB) to plug into stepper motor to run some equipment automatically.
- Help build two of the main products that are sold, using 3D-printers, a Dremel and a Mill
- Researched a CNC machine to be used in the lab to better prototype chassis for its main product.

Moog Military and Aircraft **Buffalo, NY**
Electrical engineer Inter *Aug. 2026 – May2026*

- Created an axial Flux motor using PCB as a stator instead of the traditional stator.
- Worked on creating code to be able to automate test procedures for motor controllers.
- Worked with high voltage equipment along with lab equipment to troubleshoot issues being experienced in some product testing.